

A validated structural genomics atlas of KPC carbapenemase variants across 49,774 outbreak isolates: convergent loop remodeling drives ceftazidime-avibactam resistance emergence

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Abstract

Background. *Klebsiella pneumoniae* carbapenemase (KPC) is the most widespread carbapenemase worldwide. Since the 2015 introduction of ceftazidime-avibactam (CAZ-AVI), KPC variants that escape inhibition have proliferated clinically, but a systematically validated, outbreak-scale map of KPC protein variation and its structural context has been missing. **Methods.** We built a deterministic variant caller that translates blaKPC coding sequences, aligns the protein to the KPC-2 reference, and emits variant signatures in standard Ambler numbering. The caller was validated against two independent reference databases (CARD, 231 alleles; NCBI AMRFinderPlus 2026-08-07.1, 295 alleles) with a locked 100% exact-reproduction gate before any outbreak analysis. We then censused 49,774 blaKPC-bearing *Klebsiella* isolates from NCBI Pathogen Detection (snapshot PDG000000012.2532), mapped every observed variant onto experimental KPC structures (PDB 3DW0, 4ZBE, 7TB7, 8G2R, 8AKK/8AKL), and graded resistance claims against a literature-curated set of 65 CAZ-AVI-resistant variants. **Results.** 169 distinct KPC alleles circulate in outbreak genomes (50,238 allele calls). KPC-2 (34,616 calls; 69.8%) and KPC-3 (13,788; 27.8%) dominate. 55 of 65 literature-documented CAZ-AVI-resistant alleles are observed (603 isolates), and their share of KPC⁺ isolates rose from 0% before 2013 to 3.8% in 2023. Variation is strongly convergent: 39 alleles independently carry substitutions or insertions at Ambler 179, and the omega loop (164–179), loop 237–243 and loop 266–275 account for the large majority of documented resistance events. Re-analysis of deposited structures reproduces the published omega-loop destabilization mechanism with our own numbers (omega-loop-to-chain mean B-factor ratio 0.91 in WT apo KPC-2 vs 1.14 in D179N apo; 0.96 vs 1.18 in avibactam complexes). Notably, raw proximity of a variant to the catalytic Ser70 does not distinguish resistant from other circulating alleles (median 10.6 vs 11.6 Å, Mann-Whitney $p = 0.122$); loop membership, not distance, is the discriminating feature. Eighteen outbreak-observed alleles postdate the CARD snapshot, nine of them carrying loop insertions at documented resistance positions. **Conclusions.** KPC escape from ceftazidime-avibactam follows a small set of convergent structural routes confined to three surface loops. A validated caller plus a curated structural atlas converts routine AMR allele calls into a resistance-risk annotation layer for genomic surveillance.

1 Introduction

Carbapenem-resistant Enterobacterales are a critical-priority threat, and KPC is their dominant carbapenemase worldwide. The β -lactam/ β -lactamase-inhibitor combination ceftazidime-avibactam

(CAZ-AVI), approved in 2015, restored a treatment option against KPC producers because avibactam carbamylates the catalytic Ser70 of class A enzymes. Within a few years, however, clinical KPC variants resistant to CAZ-AVI began to appear, initially as isolated case reports (e.g., KPC-31, KPC-33 carrying D179Y) and later as a steady stream of new alleles, including large insertions and deletions in the loops surrounding the active site.

Three gaps limit genomic surveillance of this evolution. First, new-variant reports are anecdotal: each paper describes one or a few alleles, usually after a treatment failure. Second, public allele reference sets disagree in coverage: the CARD snapshot we obtained carries 231 blaKPC alleles, while the current NCBI AMRFinderPlus database carries 295, and outbreak surveillance typically reports only allele names without protein-level or structural annotation. Third, no existing resource connects all three of: (i) a validated variant caller with an exact-reproduction gate, (ii) a complete census of which alleles actually circulate in outbreak genomes, and (iii) per-variant structural annotation on experimental KPC structures.

Published work covers parts of this space. An evolutionary overview curated 65 CAZ-AVI-resistant KPC variants from the literature and identified three mutational hot spots (omega loop 164–179, loop 237–243, loop 266–275). Structural studies of the D179N/D179Y variants showed omega-loop destabilization as the resistance mechanism. Laboratory selection recovered indels in the same loops. A two-year single-hospital surveillance of 4,641 isolates tracked CAZ-AVI resistance epidemiology without structural mapping. What is missing is the joint, validated view: every KPC allele circulating in public outbreak genomes, its exact protein signature in canonical Ambler numbering, and its structural context relative to substrate and inhibitor complexes.

Here we close that gap for the KPC slice of a broader carbapenemase-structure project. Contributions: (1) a deterministic KPC variant caller passing a locked 100% exact-reproduction gate on 526 allele records across two independent databases, with zero protein-level collisions; (2) a census of 169 circulating alleles across 49,774 Pathogen Detection *Klebsiella* isolates; (3) a structural effect table annotating every circulating variant with loop membership and distance to catalytic machinery; (4) quantified emergence of documented CAZ-AVI-resistant alleles from 0% to nearly 4% of KPC⁺ isolates; and (5) an independent numerical reproduction of the omega-loop destabilization mechanism from deposited structures.

2 Methods

2.1 Data sources and byte-lock

All inputs were frozen on 2026-09-23 with SHA-256 checksums recorded before analysis (full manifest in repository, `kpc/data/DATA_MANIFEST.md`). Sources: (i) CARD data tarball (2026-09-23 download), from which 231 blaKPC alleles with nucleotide and protein sequences were extracted; (ii) NCBI AMRFinderPlus database 2026-08-07.1 (`AMRProt.fa`, `AMR_CDS.fa`), 295 KPC entries; (iii) NCBI Pathogen Detection *Klebsiella* group snapshot PDG000000012.2532, AMR metadata table (190,586,905 bytes), covering all clustered *Klebsiella* isolate assemblies with AMRFinderPlus allele calls; (iv) PDB mmCIF files 2OV5, 3DW0 (WT KPC-2 apo), 4ZBE (WT KPC-2-avibactam), 8AKK/8AKL (KPC-2 E166Q acyl-enzymes with imipenem/meropenem), 7TB7 (D179N apo), 8G2R (D179N-avibactam), 8TMR (KPC-44-avibactam); (v) a literature-curated table of 65 CAZ-AVI-resistant KPC variants extracted from the evolutionary-overview review (PMC9487638), with hotspot annotations as published.

2.2 Variant caller and locked validation gate

The caller takes a blaKPC nucleotide CDS, translates it (bacterial code, table 11), globally aligns the protein to the 293-aa KPC-2 reference (pairwise alignment; match +2, mismatch -1, gap open -10, gap extend -0.5), and emits a variant signature: substitutions, insertions and deletions in KPC-2 protein coordinates. Allele assignment uses AMRFinder-style exact nucleotide identity against the reference set. Protein coordinates are converted to standard Ambler numbering through an alignment to PDB 3DW0, whose residue numbering we verified against canonical anchors (Ser70, Lys73, Glu166, Asp179, Val240, Thr243).

Gate G1 (locked before outbreak analysis): the caller must reproduce every reference allele exactly - correct translation, identical signature computed from nucleotide and from protein records, and correct exact-nt allele assignment - with 100% required and zero tolerance. Outcome: 231/231 CARD alleles and 295/295 AMRFinderPlus alleles passed all three checks; every allele is protein-distinct (zero signature collisions), so a protein-level atlas is fully resolved. A union reference set (296 alleles: CARD 231 + AMRFinderPlus 295, 230 shared) showed zero protein disagreements between the databases for shared alleles.

2.3 Outbreak census

Rows of the Pathogen Detection AMR metadata whose **AMR.genotypes** field contains a blaKPC call were retained. Calls were normalized (quote/whitespace stripping) and split into isolate-allele pairs. Isolates carry per-row metadata: assembly accession, BioSample, collection date, geographic location. Partial (**PARTIAL**, **PARTIAL_END_OF_CONTIG**) and **MISTRANSLATION** calls and unnamed blaKPC calls were counted but excluded from allele-level structural analysis.

2.4 Structural annotation

For every variant operation we recorded: Ambler position, membership in the three published resistance hot spots (omega loop 164–179; loop 237–243; loop 266–275), and minimum C α –C α distance to the catalytic Ser70 in 3DW0. Alleles were graded: *documented resistance* (in the 65-variant literature set); *consistent with resistance - hypothesis* (operation inside a hot spot or within 10 Å of Ser70, no literature evidence); *no resistance evidence*. As a structural mechanism check, we computed per-structure mean B-factors of the omega loop (164–179) relative to the whole chain for WT (3DW0, 4ZBE) and D179N (7TB7, 8G2R) structures; literature predicts loop destabilization in the D179N background.

2.5 Statistics

Distance distributions were compared with a one-sided Mann-Whitney U test ($\alpha = 0.05$). Yearly shares use collection year from isolate metadata; years after 2023 are interpreted with caution because of submission lag.

3 Results

3.1 The circulating KPC allele census

The snapshot contains 49,774 blaKPC-bearing *Klebsiella* isolates yielding 50,238 isolate-allele pairs (some isolates carry two copies). Three denominators are used consistently throughout: 49,774 isolates carry at least one blaKPC call; these yield 50,238 isolate-allele calls (some isolates carry two copies); of these, 49,589 calls resolve to a named allele across 169 distinct alleles and form the master-table denominator, while the remaining 649 calls are unnamed or partial/mistranslation (assembly-limited) and are counted but excluded from allele-level analysis. KPC-2 dominates (34,616 calls; 69.8% of named calls) followed by KPC-3 (13,788; 27.8%); the remaining 167 alleles account for 2.4% of calls (Figure 1). The long tail is not noise: it includes the documented resistance alleles KPC-33 (173 isolates), KPC-31 (118), KPC-71 (42), KPC-23 (36) and KPC-66 (22).

3.2 CAZ-AVI-resistant alleles are established and rising

Fifty-five of the 65 literature-documented CAZ-AVI-resistant alleles appear in outbreak genomes (603 isolates; Table 1). Their share of KPC⁺ isolates by collection year was 0% through 2012, first appeared in 2013, and climbed monotonically to 3.8% in 2023 (Figure 2; 2024–2026 depressed by submission lag). The rise begins after the 2015 clinical introduction of CAZ-AVI, consistent with drug-pressure selection. Geographically, China and Italy contribute the most resistant-allele isolates, matching published case-report geography, but the set spans at least 15 countries (Figure 5).

3.3 Convergent variation concentrates in three loops

Across the 169 circulating alleles, variation is strongly convergent (Figure 3). Ambler position 274 is substituted in 64 alleles, but this is the KPC-3 founder marker (H274Y), not resistance-associated. The dominant resistance-associated convergence is Ambler 179: 39 circulating alleles independently carry D179 substitutions (D179Y/N/V/A/G) or insertions at 179. Further convergent positions sit in the omega loop (164, 165, 166, 169, 170, 172, 180: 5–10 alleles each), loop 237–243 (240, 241, 242, 243: 4–8 alleles each) and loop 266–275 (261, 262, 264, 265, 268, 269: 3–9 alleles each) - exactly the three hot spots defined by the literature curation, now confirmed at outbreak scale with independent data.

3.4 Structural mechanism, independently reproduced

Re-computing B-factor statistics from deposited structures reproduces the published destabilization mechanism with our own numbers: the omega loop is *more* ordered than the chain average in WT apo KPC-2 (mean B 9.5 vs 10.5; ratio 0.91) but *less* ordered in D179N apo (11.9 vs 10.5; ratio 1.14); the same flip appears in avibactam complexes (WT 0.96 vs D179N 1.18). This is consistent with D179 anchoring the loop against the catalytic wall, and its substitution freeing the loop to distort the avibactam-binding pose.

KPC allele census, NCBI Pathogen Detection Klebsiella snapshot PDG000000012.2532 (n=49,774 is

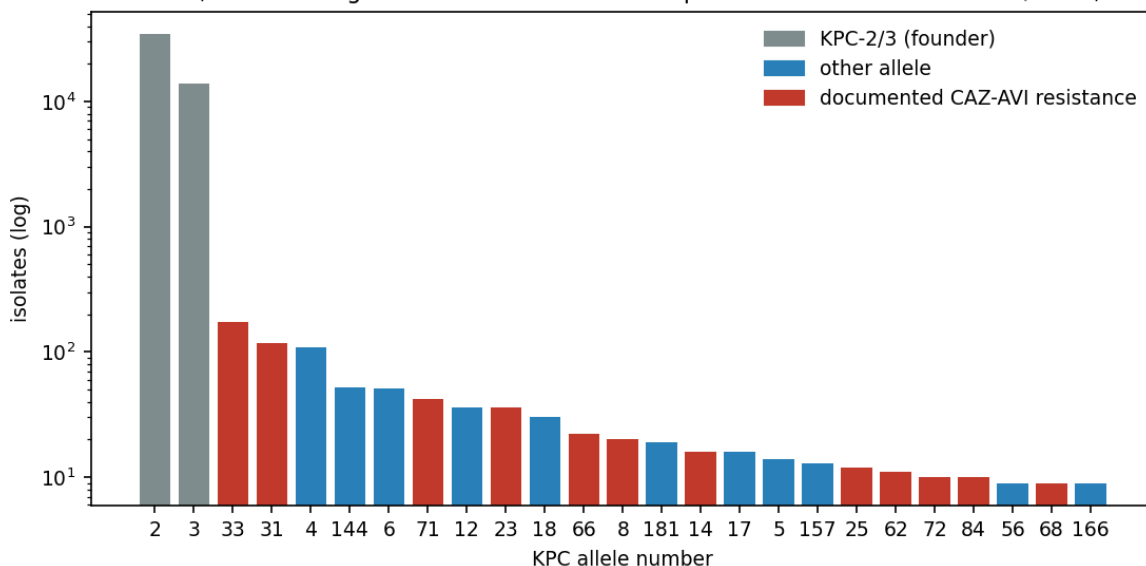


Figure 1: KPC allele census across 49,774 Pathogen Detection Klebsiella isolates (log scale). Red: documented CAZ-AVI-resistant allele.

3.5 Proximity to Ser70 does not separate resistant alleles

Against intuition, the minimum $C\alpha$ distance from a variant to Ser70 does not distinguish documented-resistant from other circulating alleles (median 10.6 Å vs 11.6 Å; Mann-Whitney $p = 0.122$; Figure 4). Loop membership is the discriminating feature: resistance variants act by reshaping the pocket-lining loops (changing ceftazidime accommodation and avibactam pose) rather than by simply sitting closer to the catalytic serine. This negative result matters for predictor design: distance-only structural scores would fail on this enzyme.

3.6 Post-CARD alleles are already in outbreaks

Eighteen circulating alleles are present in AMRFinderPlus 2026-08-07.1 but absent from the CARD snapshot (Table 2). Nine of them carry insertions or substitutions at documented resistance positions (e.g., KPC-250 D179Y+ins268_N; KPC-275 D179Y+T243P; KPC-261 D179Y+ins266_APKNKDE; KPC-272 ins165_EL+ins273_YSEAV), all collected 2021–2025. Reference-database update cadence is thus a live surveillance bottleneck: the alleles reaching patients are newer than one of the two major reference sets.

3.7 Grading summary

Of the 169 circulating alleles: 55 carry documented CAZ-AVI resistance (603 isolates); 92 are hotspot- or proximity-graded hypotheses (14,261 isolates, dominated by KPC-2/KPC-3-background alleles with single hotspot-region substitutions); the remainder have no resistance evidence. All grades, signatures and annotations are in the repository master table (296 rows).

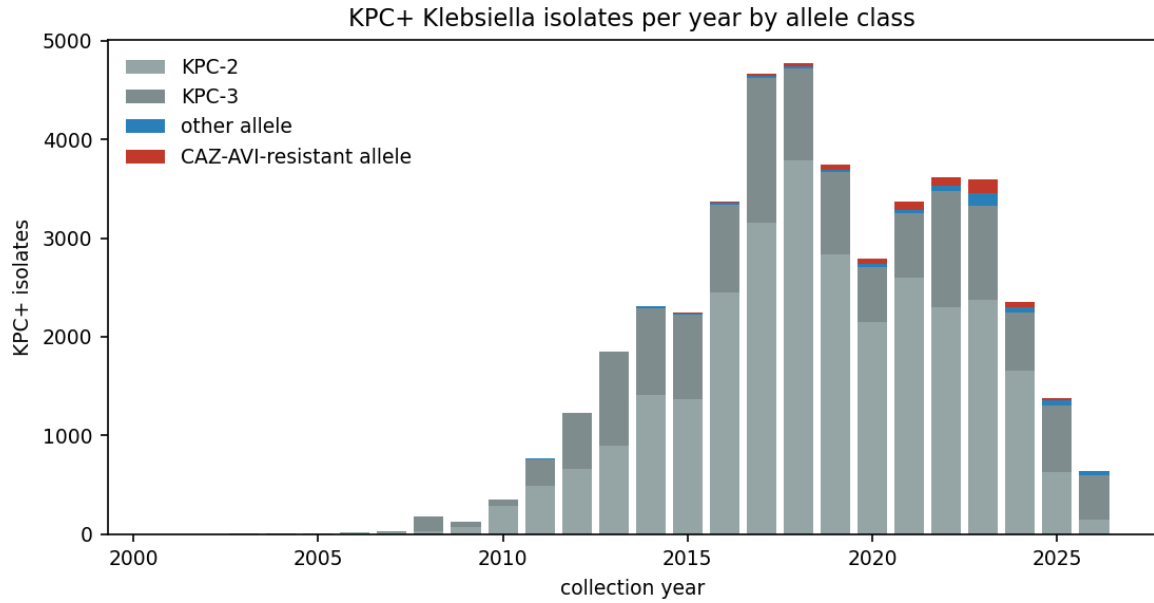


Figure 2: KPC⁺ isolates per collection year by allele class. Documented CAZ-AVI-resistant alleles (red) rise from 0% (pre-2013) to 3.8% (2023); recent years underreported.

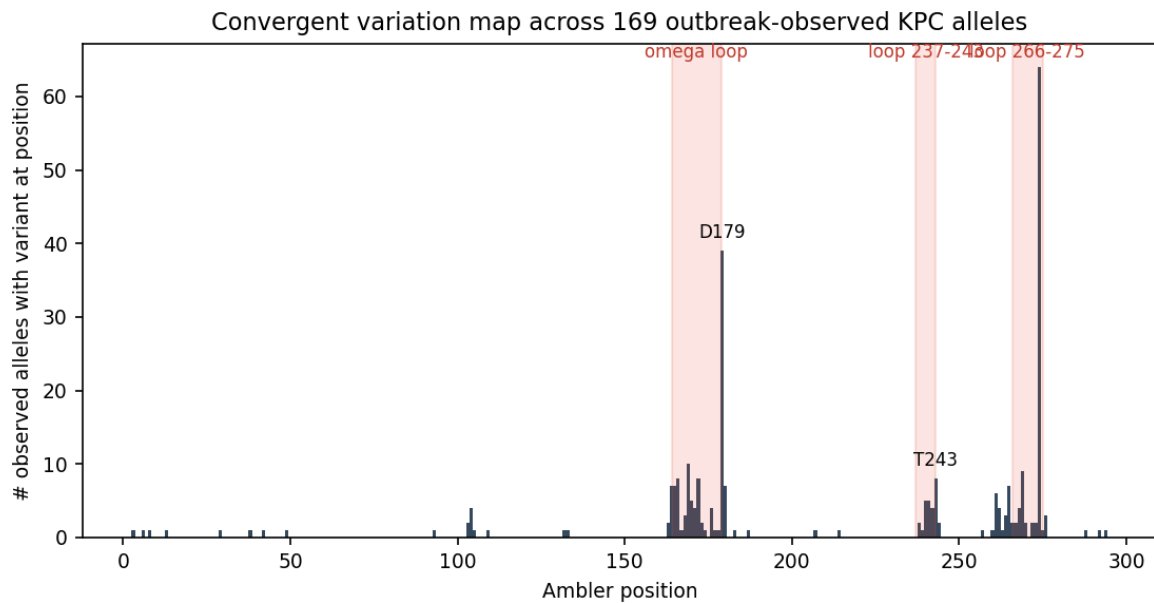


Figure 3: Convergence map: number of circulating alleles carrying a variant at each Ambler position. Red bands: the three resistance hot spots.

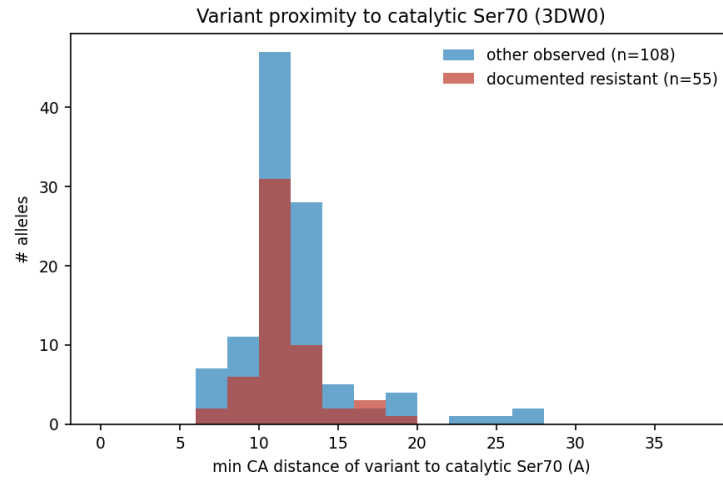


Figure 4: Minimum variant-to-Ser70 $C\alpha$ distance does not separate documented-resistant from other circulating alleles ($p = 0.122$).

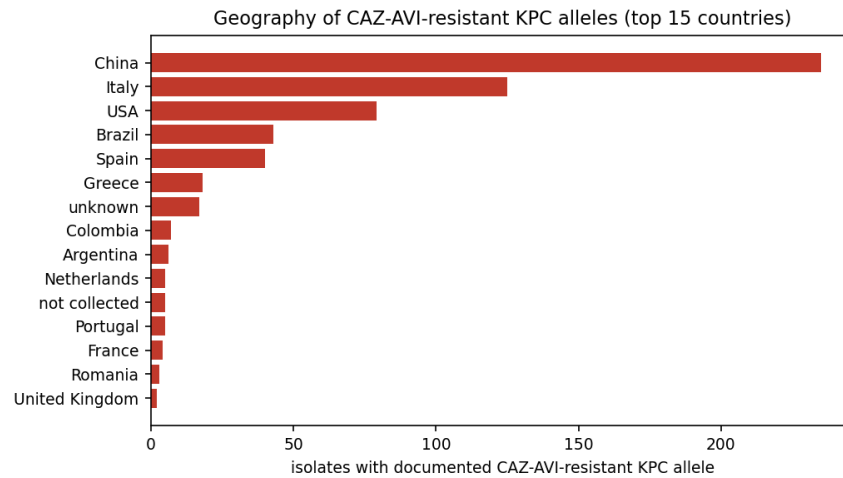


Figure 5: Countries contributing the most isolates with documented CAZ-AVI-resistant KPC alleles.

4 Discussion

Three conclusions follow. First, KPC’s escape from CAZ-AVI is not idiosyncratic: at outbreak scale it reuses a small set of structural routes, all remodeling one of three pocket-lining loops. That convergence is good news for surveillance and for inhibitor design - the escape space is bounded and enumerable, and 39-allele convergence at D179 marks the single most important watch position.

Second, validated calling matters. Our caller passed a locked 100% exact-reproduction gate on 526 allele records spanning two databases before touching outbreak data, and the two reference databases proved perfectly consistent on their 230 shared alleles. That makes the downstream atlas a measurement, not an estimate. The same discipline surfaced a real operational finding: 18 circulating alleles postdate the CARD snapshot, nine with loop insertions at documented resistance positions.

Third, structure helps where naive geometry fails. The distance-to-Ser70 negative result shows that simple proximity scores would misfire on KPC, while loop-level features (membership, indel type) and the B-factor destabilization signature reproduce the mechanism. A practical next step is a per-allele risk score combining loop membership, indel length, and loop-order proxies, calibrated on the 65-variant documented set.

Limitations. The census inherits Pathogen Detection sampling biases (country and year coverage, submission lag depressing 2024–2026). Allele calls are AMRFinderPlus-derived; the 649 unnamed/partial calls were not re-extracted from assemblies, so rare novel proteins may hide there. Resistance grading relies on a literature-curated set; alleles graded “hypothesis” require phenotypic confirmation, and MIC data are not part of the snapshot. Structural annotation uses WT and D179-background structures; allele-specific conformations (especially large indels) are approximations.

5 Data and code availability

Code, manifests with SHA-256 byte-locks, the 296-allele union reference set, the master allele table and all figures: repository `amr-carbapenem-structure`, directory `kpc/`. All sources are public (NCBI Pathogen Detection, AMRFinderPlus, CARD, RCSB PDB, PMC).

References

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Table 1: Outbreak-observed KPC alleles with documented ceftazidime-avibactam resistance (55 of 65 documented alleles observed). Ambler numbering anchored to PDB 3DW0.

Allele	Variant signature (Ambler)	Isolates	Hotspot	Top country	Years
KPC-33	D179Y	173	omega_loop	China	2016-2025
KPC-31	D179Y; H274Y	118	loop266_275;omega_loop	Italy	2014-2026
KPC-71	ins180.S	42	-	China	2017-2025
KPC-23	V240A; H274Y	36	loop237_243;loop266_275	Greece	2014-2025
KPC-66	del166.EL; H274Y	22	loop266_275;omega_loop	Italy	2019-2024
KPC-8	V240G; H274Y	20	loop237_243;loop266_275	USA	2013-2021
KPC-14	del242.GT	16	loop237_243	China	2003-2024
KPC-25	ins165.EL	12	omega_loop	China	2020-2025
KPC-62	L169Q; H274Y	11	loop266_275;omega_loop	Spain	2019-2023
KPC-72	A172D	10	omega_loop	China	2017-2023
KPC-84	T243P	10	loop237_243	China	2016-2025
KPC-68	ins180.SS; H274Y	9	loop266_275	USA	2019-2026
KPC-44	ins261.AVYTRAPNKDDKHSE	8	-	China	2020-2024
KPC-90	ins179.TY	8	omega_loop	China	2018-2025
KPC-46	L169P; H274Y	7	loop266_275;omega_loop	Portugal	2019-2024
KPC-49	R164S; H274Y	7	loop266_275;omega_loop	Italy	2018-2024
KPC-53	ins165.EL; H274Y	7	loop266_275;omega_loop	Italy	2016-2026
KPC-29	ins269.KDD; H274Y	6	loop266_275	USA	2019-2024
KPC-35	L169P	6	omega_loop	China	2020-2022
KPC-78	D179A	6	omega_loop	China	2019-2023
KPC-39	A172T; H274Y	5	loop266_275;omega_loop	France	2018-2024
KPC-88	D176Y	5	omega_loop	China	2017-2023
KPC-65	ins179.TY; H274Y	4	loop266_275;omega_loop	USA	2017-2023
KPC-86	D179G	4	omega_loop	China	2022-2023
KPC-50	H274Y; ins275.EAV	3	loop266_275	USA	2024-2025
KPC-57	D179V	3	omega_loop	China	2020-2022
KPC-67	ins269.KDDKDD; H274Y	3	loop266_275	Italy	2019-2020
KPC-76	D179Y; ins262.VYTRAPN	3	omega_loop	China	2019-2022
KPC-85	A172V; H274Y	3	loop266_275;omega_loop	Netherlands	2018-2021
KPC-94	del169.L; N170H; H274Y	3	loop266_275;omega_loop	Italy	2017-2024
KPC-32	D179Y; T243M; H274Y	2	loop237_243;loop266_275;omega_loop	USA	2015-2015
KPC-36	D163E; H274Y	2	loop266_275	Italy	2017-2019
KPC-41	ins267.PNK; H274Y	2	loop266_275	not collected	2018-2018
KPC-74	del239.GV	2	loop237_243	China	2020-2025
KPC-79	ins262.VYTRAPN	2	-	China	2019-2022
KPC-103	ins269.KDDKHSEAVIAA	2	loop266_275	Brazil	2019-2019
KPC-112	del166.EL; del242.GT	2	loop237_243;omega_loop	China	2023-2024
KPC-114	ins180.SS	2	-	unknown	2020-2023
KPC-51	D179N; Y241H; H274N	1	loop237_243;loop266_275;omega_loop	Iran	2018-2018
KPC-52	D179Y; ins262.V	1	omega_loop	China	2018-2018
KPC-58	ins265.RAPNKDDN	1	-	Croatia	2019-2019
KPC-69	ins165.GL; H274Y	1	loop266_275;omega_loop	Italy	2020-2020
KPC-70	D179Y; T265A; H274Y	1	loop266_275;omega_loop	Italy	2020-2020
KPC-80	ins267.PNK	1	loop266_275	Argentina	2019-2019
KPC-95	A172T; D179Y; H274Y	1	loop266_275;omega_loop	Spain	2018-2018
KPC-96	Y241N	1	loop237_243	Argentina	2019-2019
KPC-97	ins274.SEAVN	1	loop266_275	Argentina	2020-2020
KPC-98	R164H; H274Y	1	loop266_275;omega_loop	Italy	2023-2023
KPC-104	ins179.TY; ins270.DDKHSE	1	loop266_275;omega_loop	Brazil	2020-2020
KPC-105	L169Q; ins261.AVYTRAPNKDDKHSE	1	omega_loop	Brazil	2020-2020
KPC-106	ins179.TY; ins274.SEAV	1	loop266_275;omega_loop	Brazil	2021-2021
KPC-107	ins180.SSPRAVTESLQKLTLSALAAPQRQQFV			Brazil	2020-2020

continued

Table 1: Outbreak-observed KPC alleles with documented ceftazidime-avibactam resistance (55 of 65 documented alleles observed). Ambler numbering anchored to PDB 3DW0.

Allele	Variant signature (Ambler)	Isolates	Hotspot	Top country	Years
KPC-108	ins179_TSSPRAVTEN; ins261_AVYTRAPNKDDKHSE	1	omega_loop	Brazil	2021-2021
KPC-113	ins266_G	1	loop266_275	Brazil	2020-2020
KPC-115	del169_LN; S171P; H274Y	1	loop266_275;omega_loop	Argentina	2021-2021

Table 2: Outbreak-observed alleles present in AMRFinderPlus 2026-08-07.1 but absent from the CARD 2026-09-23 snapshot (post-CARD named alleles).

Allele	Variant signature (Ambler)	Isolates	Top country	Years
KPC-244	N214T	3	USA	2016-2019
KPC-250	D179Y; ins268_N	2	China	2023-2023
KPC-262	P268S	2	China	2022-2024
KPC-264	S109P	2	China	2016-2025
KPC-275	D179Y; T243P	2	China	2025-2025
KPC-276	T243A; H274Y	2	USA	2015-2018
KPC-245	ins263_YTRAPNKDD; H274Y; E276D	1	Italy	2024-2024
KPC-252	ins243_ANDYAA	1	China	2022-2022
KPC-259	ins264_TRAPN	1	USA	2017-2017
KPC-261	D179Y; ins266_APNKDE	1	China	2021-2021
KPC-263	ins238_G	1	China	2018-2018
KPC-272	ins165_EL; ins273_YSEAV; H274N	1	unknown	2025-2025
KPC-281	A257S	1	Slovakia	2025-2025
KPC-282	ins264_TRAPNK	1	Italy	2023-2023
KPC-287	L169Q; del242_GT	1	China	2025-2025
KPC-295	ins165_EL; T243A; H274Y	1	Italy	2022-2022
KPC-297	ins168_LK; del242_GT	1	China	2025-2025
KPC-300	P268T; H274R	1	China	2022-2022

A Full outbreak census of 169 observed KPC alleles

Table 3: Complete census of the 169 outbreak-observed KPC alleles (snapshot PDG000000012.2532). Distance = minimum C α distance of any variant to Ser70 (3DW0).

Allele	Signature (Ambler)	Isolates	Dist (Å)	Grade	Top country	Years
KPC-2	identical to KPC-2	34616	-	no evidence	China	1996-2026
KPC-3	H274Y	13788	13.9	hypothesis	USA	2001-2026
KPC-33	D179Y	173	10.0	documented	China	2016-2025
KPC-31	D179Y; H274Y	118	10.0	documented	Italy	2014-2026
KPC-4	P104R; V240G	108	12.9	hypothesis	USA	2011-2026
KPC-144	A172T	52	11.5	hypothesis	China	2018-2023
KPC-6	V240G	51	12.9	hypothesis	USA	2014-2026
KPC-71	ins180_S	42	10.9	documented	China	2017-2025
KPC-12	L169M	36	8.8	hypothesis	China	2014-2024
KPC-23	V240A; H274Y	36	12.9	documented	Greece	2014-2025
KPC-18	V8I	30	-	no evidence	USA	2016-2026
KPC-66	del166_EL; H274Y	22	8.9	documented	Italy	2019-2024
KPC-8	V240G; H274Y	20	12.9	documented	USA	2013-2021
KPC-181	E276D	19	14.7	no evidence	Brazil	2016-2025
KPC-14	del242_GT	16	10.9	documented	China	2003-2024
KPC-17	F207L	16	16.2	no evidence	Taiwan	2014-2024
KPC-5	P104R	14	13.1	no evidence	USA	2014-2026
KPC-157	N132S	13	8.4	hypothesis	China	2016-2024
KPC-25	ins165_EL	12	10.6	documented	China	2020-2025
KPC-62	L169Q; H274Y	11	8.8	documented	Spain	2019-2023
KPC-72	A172D	10	11.5	documented	China	2017-2023
KPC-84	T243P	10	10.9	documented	China	2016-2025
KPC-56	H274Y; G294W	9	13.9	hypothesis	USA	2013-2022
KPC-68	ins180_SS; H274Y	9	10.9	documented	USA	2019-2026
KPC-166	del169_L; N170H	9	7.9	hypothesis	Italy	2014-2022
KPC-170	D179N	9	10.0	hypothesis	China	2017-2023
KPC-173	ins168_LK	9	11.4	hypothesis	China	2016-2023
KPC-44	ins261_AVYTRAPNKDDKHSE	8	16.6	documented	China	2020-2024
KPC-90	ins179_TY	8	10.0	documented	China	2018-2025
KPC-93	ins265_RAPNN	8	13.2	no evidence	China	2021-2023
KPC-184	H274Y; E276D	8	13.9	hypothesis	Italy	2019-2026
KPC-10	P104R; H274Y	7	13.1	hypothesis	USA	2014-2025
KPC-46	L169P; H274Y	7	8.8	documented	Portugal	2019-2024
KPC-49	R164S; H274Y	7	11.9	documented	Italy	2018-2024
KPC-53	ins165_EL; H274Y	7	10.6	documented	Italy	2016-2026
KPC-148	ins261_AVYTRAPNKDDKYSE; H274Y	7	13.9	hypothesis	USA	2015-2024
KPC-176	S171P	7	11.6	hypothesis	China	2017-2024
KPC-29	ins269_KDD; H274Y	6	13.9	documented	USA	2019-2024
KPC-35	L169P	6	8.8	documented	China	2020-2022
KPC-78	D179A	6	10.0	documented	China	2019-2023
KPC-204	ins269_KDD	6	19.9	hypothesis	China	2017-2023
KPC-227	D179Y; ins269_KDD	6	10.0	hypothesis	China	2023-2023
KPC-39	A172T; H274Y	5	11.5	documented	France	2018-2024
KPC-88	D176Y	5	16.6	documented	China	2017-2023
KPC-130	D179G; H274Y	5	10.0	hypothesis	Italy	2020-2026
KPC-160	del166_EL	5	8.9	hypothesis	China	2021-2024
KPC-212	R164L	5	11.9	hypothesis	China	2023-2023
KPC-27	W105R; H274Y	4	11.0	hypothesis	Spain	2013-2014
KPC-65	ins179_TY; H274Y	4	10.0	documented	USA	2017-2023

continued

Table 3: Complete census of the 169 outbreak-observed KPC alleles (snapshot PDG000000012.2532). Distance = minimum C α distance of any variant to Ser70 (3DW0).

Allele	Signature (Ambler)	Isolates	Dist (Å)	Grade	Top country	Years
KPC-86	D179G	4	10.0	documented	China	2022-2023
KPC-126	A172V	4	11.5	hypothesis	China	2020-2023
KPC-145	D179Y; T265A	4	10.0	hypothesis	China	2023-2024
KPC-167	D179Y; ins270_DDKYSE; H274Y	4	10.0	hypothesis	Italy	2021-2022
KPC-168	D176N	4	16.6	hypothesis	China	2016-2023
KPC-38	H274Y; V292A	3	13.9	hypothesis	USA	2015-2025
KPC-50	H274Y; ins275_EAV	3	13.9	documented	USA	2024-2025
KPC-57	D179V	3	10.0	documented	China	2020-2022
KPC-61	S171P; H274Y	3	11.6	hypothesis	Brazil	2020-2020
KPC-67	ins269_KDDKDD; H274Y	3	13.9	documented	Italy	2019-2020
KPC-76	D179Y; ins262_VYTRAPN	3	10.0	documented	China	2019-2022
KPC-85	A172V; H274Y	3	11.5	documented	Netherlands	2018-2021
KPC-92	del166_EL; N170D; H274Y	3	7.9	hypothesis	Spain	2020-2020
KPC-94	del169_L; N170H; H274Y	3	7.9	documented	Italy	2017-2024
KPC-100	T243M	3	10.9	hypothesis	China	2022-2023
KPC-121	ins180_S; H274Y	3	10.9	hypothesis	Italy	2021-2023
KPC-154	ins265_RAPNKDDKYS; H274Y	3	13.2	hypothesis	Italy	2022-2022
KPC-172	D179E	3	10.0	hypothesis	China	2016-2017
KPC-205	ins265_RAPNN; H274Y	3	13.2	hypothesis	Italy	2022-2025
KPC-239	del240_V; Y241D; H274Y	3	12.9	hypothesis	Italy	2022-2023
KPC-244	N214T	3	12.6	no evidence	USA	2016-2019
KPC-30	R6H	2	-	no evidence	Brazil	2016-2016
KPC-32	D179Y; T243M; H274Y	2	10.0	documented	USA	2015-2015
KPC-36	D163E; H274Y	2	13.2	documented	Italy	2017-2019
KPC-41	ins267_PNK; H274Y	2	13.9	documented	not collected	2018-2018
KPC-74	del239_GV	2	10.6	documented	China	2020-2025
KPC-75	L13F	2	-	no evidence	China	2019-2019
KPC-79	ins262_VYTRAPN	2	13.6	documented	China	2019-2022
KPC-81	del173_I	2	15.3	hypothesis	Argentina	2020-2020
KPC-99	R164S	2	11.9	hypothesis	Romania	2020-2023
KPC-103	ins269_KDDKHSEAVIAA	2	19.9	documented	Brazil	2019-2019
KPC-112	del166_EL; del242_GT	2	8.9	documented	China	2023-2024
KPC-114	ins180_SS	2	10.9	documented	unknown	2020-2023
KPC-118	V103L; W165R; H274Y	2	10.6	hypothesis	USA	2020-2021
KPC-124	del166_EL; D179Y; H274Y	2	8.9	hypothesis	Portugal	2022-2022
KPC-125	D179A; H274Y	2	10.0	hypothesis	Italy	2021-2025
KPC-127	ins165_EL; A172T	2	10.6	hypothesis	China	2022-2025
KPC-141	D179Y; ins238_G	2	7.3	hypothesis	Brazil	2021-2022
KPC-179	A133T; ins180_S	2	10.9	no evidence	China	2023-2023
KPC-180	R164H	2	11.9	hypothesis	China	2022-2023
KPC-190	D179Y; A244V	2	8.9	hypothesis	China	2024-2024
KPC-203	del166_EL; ins260_LAVYTRAPM; H274Y	2	8.9	hypothesis	Italy	2022-2022
KPC-216	ins169_K; H274Y	2	8.8	hypothesis	Italy	2022-2022
KPC-250	D179Y; ins268_N	2	10.0	hypothesis	China	2023-2023
KPC-262	P268S	2	19.7	hypothesis	China	2022-2024
KPC-264	S109P	2	15.7	no evidence	China	2016-2025
KPC-275	D179Y; T243P	2	10.0	hypothesis	China	2025-2025
KPC-276	T243A; H274Y	2	10.9	hypothesis	USA	2015-2018
KPC-7	M49I; H274Y	1	13.9	hypothesis	unknown	-
KPC-11	P104L	1	13.1	no evidence	unknown	-
KPC-20	V29I	1	-	no evidence	USA	2017-2017

continued

Table 3: Complete census of the 169 outbreak-observed KPC alleles (snapshot PDG000000012.2532). Distance = minimum $C\alpha$ distance of any variant to Ser70 (3DW0).

Allele	Signature (Ambler)	Isolates	Dist (Å)	Grade	Top country	Years
KPC-45	T93K	1	23.6	no evidence	USA	2020-2020
KPC-51	D179N; Y241H; H274N	1	10.0	documented	China	2018-2018
KPC-52	D179Y; ins262_V	1	10.0	documented	China	2018-2018
KPC-58	ins265_RAPNKDDN	1	13.2	documented	Croatia	2019-2019
KPC-69	ins165_GL; H274Y	1	10.6	documented	Italy	2020-2020
KPC-70	D179Y; T265A; H274Y	1	10.0	documented	Italy	2020-2020
KPC-77	R164P	1	11.9	hypothesis	China	2022-2022
KPC-80	ins267_PNK	1	17.2	documented	Argentina	2019-2019
KPC-91	V103L; H274Y	1	13.9	hypothesis	Taiwan	2020-2020
KPC-95	A172T; D179Y; H274Y	1	10.0	documented	Spain	2018-2018
KPC-96	Y241N	1	14.6	documented	Argentina	2019-2019
KPC-97	ins274_SEAVN	1	13.9	documented	Argentina	2020-2020
KPC-98	R164H; H274Y	1	11.9	documented	Italy	2023-2023
KPC-101	ins272_KH	1	18.2	hypothesis	China	2023-2023
KPC-104	ins179_TY; ins270_DDKHSE	1	10.0	documented	Brazil	2020-2020
KPC-105	L169Q; ins261_AVYTRAPNKDDKHSE	1	8.8	documented	Brazil	2020-2020
KPC-106	ins179_TY; ins274_SEAV	1	10.0	documented	Brazil	2021-2021
KPC-107	ins180_SSPRAVTESLQKLTLSALAAPQRQQFW	1	10.9	documented	Brazil	2020-2020
KPC-108	ins179_TSSPRAVTEN; ins261_AVYTRAPNKDDKHSE	1	10.0	documented	Brazil	2021-2021
KPC-109	ins268_NKDDKY; H274Y	1	13.9	hypothesis	Italy	2021-2021
KPC-110	G42R; D179Y; H274Y	1	10.0	hypothesis	Italy	2021-2021
KPC-113	ins266_G	1	14.1	documented	Brazil	2020-2020
KPC-115	del169_LN; S171P; H274Y	1	7.9	documented	Argentina	2021-2021
KPC-119	Q38H	1	25.3	no evidence	USA	2018-2018
KPC-128	D179Y; T243M	1	10.0	hypothesis	China	2022-2022
KPC-129	N170H; ins262_VYTRAPNK	1	7.9	hypothesis	China	2022-2022
KPC-132	ins265_RAPNKDDKS; H274Y	1	13.2	hypothesis	Spain	2022-2022
KPC-135	del166_EL; ins261_AVYTRAPNKDDKHSE	1	8.9	hypothesis	China	2023-2023
KPC-139	D179Y; ins269_KDDKHS	1	10.0	hypothesis	Brazil	2020-2020
KPC-140	D179N; ins261_AVYTRAPNKDDKHSE	1	10.0	hypothesis	Brazil	2021-2021
KPC-142	ins180_S; ins272_KHSEAV	1	10.9	hypothesis	Brazil	2021-2021
KPC-143	D179Y; T187S	1	10.0	hypothesis	Brazil	2021-2021
KPC-146	L3M	1	-	no evidence	China	-
KPC-150	del167_LEL; N170H; H274Y	1	7.9	hypothesis	Spain	-
KPC-158	A244V	1	8.9	hypothesis	China	2017-2017
KPC-162	ins264_TRAPNKD	1	11.9	no evidence	Argentina	2022-2022
KPC-163	ins269_KDDKHS	1	19.9	hypothesis	Argentina	2022-2022
KPC-169	del176_DAR	1	13.6	hypothesis	China	2017-2017
KPC-171	ins274_SEAVIAAD	1	13.9	hypothesis	China	2017-2017
KPC-174	del241_Y	1	14.6	hypothesis	China	2019-2019
KPC-175	ins179_T	1	10.0	hypothesis	China	2019-2019
KPC-177	ins179_TSSPRAVTESLQN	1	10.0	hypothesis	China	2016-2016
KPC-178	P174L; H274Y	1	13.9	hypothesis	USA	2018-2018
KPC-183	ins269_KDDKYS; H274Y	1	13.9	hypothesis	Italy	2019-2019
KPC-189	A172T; D179Y	1	10.0	hypothesis	China	2023-2023
KPC-194	D179Y; P183L	1	10.0	hypothesis	China	2023-2023
KPC-195	E288D	1	26.8	no evidence	USA	2011-2011
KPC-197	del166_EL; A177E; ins273_YS; H274D	1	8.9	hypothesis	Colombia	2023-2023

continued

Table 3: Complete census of the 169 outbreak-observed KPC alleles (snapshot PDG000000012.2532). Distance = minimum C α distance of any variant to Ser70 (3DW0).

Allele	Signature (Ambler)	Isolates	Dist (Å)	Grade	Top country	Years
KPC-202	ins274_SEAVIAAAAN	1	13.9	hypothesis	China	2023-2023
KPC-211	del176_DARD	1	10.0	hypothesis	China	2023-2023
KPC-213	D163G	1	13.2	no evidence	China	2023-2023
KPC-214	R178H	1	13.6	hypothesis	China	2023-2023
KPC-215	I173S	1	15.3	hypothesis	China	2023-2023
KPC-228	del168_ELNS	1	7.9	hypothesis	China	2021-2021
KPC-230	R164S; ins269_KDDKHS	1	11.9	hypothesis	China	2024-2024
KPC-231	S171Y; H274Y	1	11.6	hypothesis	Italy	2022-2022
KPC-234	D179G; Y241D	1	10.0	hypothesis	China	2023-2023
KPC-245	ins263_YTRAPNKDD; H274Y; E276D	1	13.5	hypothesis	Italy	2024-2024
KPC-252	ins243_ANDYAA	1	10.9	hypothesis	China	2022-2022
KPC-259	ins264_TRAPN	1	11.9	no evidence	USA	2017-2017
KPC-261	D179Y; ins266_APNKDE	1	10.0	hypothesis	China	2021-2021
KPC-263	ins238_G	1	7.3	hypothesis	China	2018-2018
KPC-272	ins165_EL; ins273_YSEAV; H274N	1	10.6	hypothesis	unknown	2025-2025
KPC-281	A257S	1	27.5	no evidence	Slovakia	2025-2025
KPC-282	ins264_TRAPNK	1	11.9	no evidence	Italy	2023-2023
KPC-287	L169Q; del242_GT	1	8.8	hypothesis	China	2025-2025
KPC-295	ins165_EL; T243A; H274Y	1	10.6	hypothesis	Italy	2022-2022
KPC-297	ins168_LK; del242_GT	1	10.9	hypothesis	China	2025-2025
KPC-300	P268T; H274R	1	13.9	hypothesis	China	2022-2022

B Convergent Ambler positions across circulating alleles

Table 4: Convergent Ambler positions: all positions varied in at least two independent circulating alleles.

Ambler position	# alleles	Alleles
274	64	KPC-3, KPC-7, KPC-8, KPC-10, KPC-23, KPC-27, KPC-29, KPC-31, KPC-32, KPC-36, KPC-38, KPC-39, KPC-41, KPC-46, KPC-49, KPC-50, KPC-51, KPC-53, KPC-56, KPC-61, KPC-62, KPC-65, KPC-66, KPC-67, KPC-68, KPC-69, KPC-70, KPC-85, KPC-91, KPC-92, KPC-94, KPC-95, KPC-97, KPC-98, KPC-106, KPC-109, KPC-110, KPC-115, KPC-118, KPC-121, KPC-124, KPC-125, KPC-130, KPC-132, KPC-148, KPC-150, KPC-154, KPC-167, KPC-171, KPC-178, KPC-183, KPC-184, KPC-197, KPC-202, KPC-203, KPC-205, KPC-216, KPC-231, KPC-239, KPC-245, KPC-272, KPC-276, KPC-295, KPC-300
179	39	KPC-31, KPC-32, KPC-33, KPC-51, KPC-52, KPC-57, KPC-65, KPC-70, KPC-76, KPC-78, KPC-86, KPC-90, KPC-95, KPC-104, KPC-106, KPC-108, KPC-110, KPC-124, KPC-125, KPC-128, KPC-130, KPC-139, KPC-140, KPC-141, KPC-143, KPC-145, KPC-167, KPC-170, KPC-172, KPC-175, KPC-177, KPC-189, KPC-190, KPC-194, KPC-227, KPC-234, KPC-250, KPC-261, KPC-275
169	10	KPC-12, KPC-35, KPC-46, KPC-62, KPC-94, KPC-105, KPC-115, KPC-166, KPC-216, KPC-287
269	9	KPC-29, KPC-67, KPC-103, KPC-139, KPC-163, KPC-183, KPC-204, KPC-227, KPC-230
172	8	KPC-39, KPC-72, KPC-85, KPC-95, KPC-126, KPC-127, KPC-144, KPC-189
166	8	KPC-66, KPC-92, KPC-112, KPC-124, KPC-135, KPC-160, KPC-197, KPC-203
243	8	KPC-32, KPC-84, KPC-100, KPC-128, KPC-252, KPC-275, KPC-276, KPC-295
180	7	KPC-68, KPC-71, KPC-107, KPC-114, KPC-121, KPC-142, KPC-179
165	7	KPC-25, KPC-53, KPC-69, KPC-118, KPC-127, KPC-272, KPC-295
265	7	KPC-58, KPC-70, KPC-93, KPC-132, KPC-145, KPC-154, KPC-205
164	7	KPC-49, KPC-77, KPC-98, KPC-99, KPC-180, KPC-212, KPC-230
261	6	KPC-44, KPC-105, KPC-108, KPC-135, KPC-140, KPC-148
240	5	KPC-4, KPC-6, KPC-8, KPC-23, KPC-239
170	5	KPC-92, KPC-94, KPC-129, KPC-150, KPC-166
241	5	KPC-51, KPC-96, KPC-174, KPC-234, KPC-239
104	4	KPC-4, KPC-5, KPC-10, KPC-11
242	4	KPC-14, KPC-112, KPC-287, KPC-297
171	4	KPC-61, KPC-115, KPC-176, KPC-231
176	4	KPC-88, KPC-168, KPC-169, KPC-211
262	4	KPC-52, KPC-76, KPC-79, KPC-129

continued

Table 4: Convergent Ambler positions: all positions varied in at least two independent circulating alleles.

Ambler position	# alleles	Alleles
268	4	KPC-109, KPC-250, KPC-262, KPC-300
276	3	KPC-181, KPC-184, KPC-245
168	3	KPC-173, KPC-228, KPC-297
264	3	KPC-162, KPC-259, KPC-282
270	2	KPC-104, KPC-167
163	2	KPC-36, KPC-213
267	2	KPC-41, KPC-80
173	2	KPC-81, KPC-215
103	2	KPC-91, KPC-118
238	2	KPC-141, KPC-263
244	2	KPC-158, KPC-190
272	2	KPC-101, KPC-142
266	2	KPC-113, KPC-261
273	2	KPC-197, KPC-272

C Reproducibility

Environment. Ubuntu 22.04 container, Python 3.10.12, Biopython 1.88, pandas, numpy, matplotlib, scipy; pandoc/pdflatex for typesetting. No GPU; all analyses CPU-bound and complete in seconds. **Pipeline.** (1) `code/kpc_caller.py` - variant caller and Gate G1 (CARD 231/231). (2) `code/extend_ref.py` - union reference build and Gate G1-extended (AMRFinderPlus 295/295), cross-database consistency. (3) `code/build_master.py` - Ambler mapping via 3DW0, structural annotation, literature grading, census join; writes `results/master_allele_table.ts`. (4) `code/figures.py` - Figures 1-5 and the distance statistics. All randomness-free and deterministic. **Byte-lock.** Every bulk input carries a SHA-256 recorded 2026-09-23 before analysis in `data/DATA_MANIFEST.md`; the Pathogen Detection snapshot id (PDG000000012.2532) pins the census. **Gate provenance.** `GATES_LOCKED.md` records the locked gates; they were reported to the orchestrator before any download or analysis (13:29 IST, 2026-09-23).